

information, since it also depends upon position of power planes, via hole diameter, plating thickness, etc. It should be kept in mind that a single via is going to add 1-3nH of inductance in the signal. Further study is required if a designer comes across such a situation. This is known as Black Magic over a PCB when compared with circuit simulation results.

Current carrying capacity of a via

Generally, I never recommend switching a power rail between the layers when designing a power PCB. However, one can use component-mounting pads for the same. If it becomes necessary, then use multiple via holes.

As per researchers, calculating current carrying capacity of a via hole is a function of multiple fac-

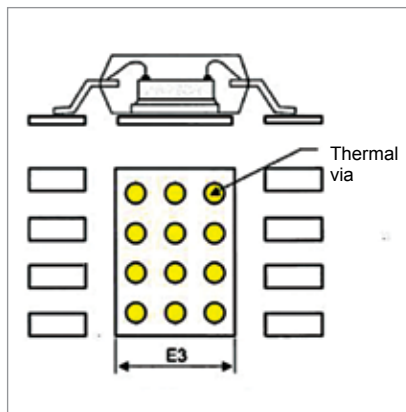


Fig. 4: Thermal vias for a package with back-surface heat dissipation

tors like, length of via, its drill diameter, placement of power planes, amount of power to be switched between the layers, glass transition temperature (T_g) of PCB base material, copper plating thickness, etc. As a thumb rule the finished diameter of a via hole must be at least

as large as the trace width divided by three. For a digital circuit, the use of via holes for V_{CC} or Ground is obvious since the amount of power is very less.

Planning via holes diameter for your design is of great concern with respect to a manufacturer's capabilities. You may plan different via hole diameter for a PCB as per your requirement.

Most common drill diameter of a via hole is 0.5mm (20mils), which can be drilled by most PCB manufacturers. If you are planning anything less than this, you must check the manufacturer's drilling and plating capabilities.

With the reduction in drill diameter, the spindle speed of a CNC PCB drilling machine needs to be high. That's why micro vias are laser drilled instead of mechanical drilling. **EFY**

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